

MICROBIOLOGY – SHORT EXAM NOTES

Culture Media • Culture Methods incl. Anaerobic Methods • Biochemical Tests (Ananthanarayan & Paniker, Ch. 4–6)

1. CULTURE MEDIA

A **culture medium** is a mixture of nutrients that supports the growth of microorganisms; it must contain a source of **carbon and energy**. Isolation on media is the starting point for **identification and antibiotic susceptibility testing**.

Common ingredients: **Agar** – from seaweed, no nutritive value, not affected by bacterial growth; **melts at 98°C, sets at 42°C**; 2% used for solid media (0.5% for semisolid). **Peptone** – partially digested protein (proteoses, polypeptides, amino acids, salts, growth factors). Others: meat extract, yeast extract, blood, serum, salts.

Liquid vs Solid Media

Liquid media (broths) give **diffuse growth**; used for large-volume samples (blood, water), enrichment and bulk cultures for antigens/vaccines, but cannot separate mixed growth. **Solid media** allow formation of **colonies** (each a clone from one cell), showing colony morphology, pigment and haemolysis, and help in **isolation of pure cultures**.

Classification of Media

Type	Definition / purpose	Examples
Simple (basal)	Basic nutrients	Nutrient broth (peptone, meat extract, NaCl, water); nutrient agar (+2% agar)
Complex	Added ingredients for special purposes	Most media, e.g. TCBS
Synthetic (defined)	Pure chemicals of known composition; research	Hank's balanced salt solution
Enriched	Basal medium + blood, serum or egg for fastidious organisms	Blood agar, chocolate agar , BHI broth, Todd–Hewitt broth
Enrichment (liquid)	Stimulate the pathogen and inhibit commensals in mixed-flora samples	Selenite F broth, tetrathionate broth (feces)
Selective (solid)	Inhibitory substance in solid medium; pathogen forms more colonies	DCA, TCBS, SS agar, Thayer–Martin, L–J medium
Indicator	Indicator changes colour when the organism grows	Wilson–Blair (<i>S. Typhi</i> – black), Potassium tellurite (McLeod's) (<i>C. diphtheriae</i> – black)
Differential	Distinguish groups by a property	MacConkey (LF pink/NLF pale), Mannitol salt agar , blood agar (haemolysis)
Transport	Preserve delicate organisms in transit	Stuart's (gonococci), buffered glycerol saline (enteric bacilli), Cary–Blair
Anaerobic	Reducing substances / absence of O ₂	Robertson's cooked meat, thioglycollate broth
Fungal	Low pH (~5.0), high sugar	Sabouraud dextrose agar (± chloramphenicol, gentamicin)
Special use		Mueller–Hinton agar (AST), screen agars (MRSA), CHROM agar (<i>Candida</i>)

Important Media

- **Blood agar** – nutrient agar + **5–10% sheep blood**; enriched and differential. **β -haemolysis** – complete lysis, clear zone (*S. pyogenes*); **α -haemolysis** – partial, **greenish** zone (viridans streptococci); **no (γ) haemolysis** – enterococci.
- **Chocolate agar** – blood agar **heated**; releases **X factor (haemin)** and **V factor (NAD)**; for *H. influenzae*, *N. meningitidis*, *N. gonorrhoeae*, pneumococcus.
- **MacConkey agar** – peptone, **lactose**, agar, **neutral red** (indicator), **bile salt (taurocholate)**; **lactose fermenters pink** (*E. coli*, *Klebsiella* – mucoid), **non-lactose fermenters pale** (*Salmonella*, *Shigella*).
- **TCBS agar** – alkaline, selective for *Vibrio*; sucrose fermenters (*V. cholerae*) **yellow**, non-fermenters (*V. parahaemolyticus*) **green**.
- **Thayer–Martin** – chocolate agar + **vancomycin, colistin, nystatin**; *N. gonorrhoeae* in 5–10% CO₂.
- **Löwenstein–Jensen (L–J)** – mineral salts, asparagine, glycerol, **malachite green, egg**; sterilised by **inspissation**; for **mycobacteria**.

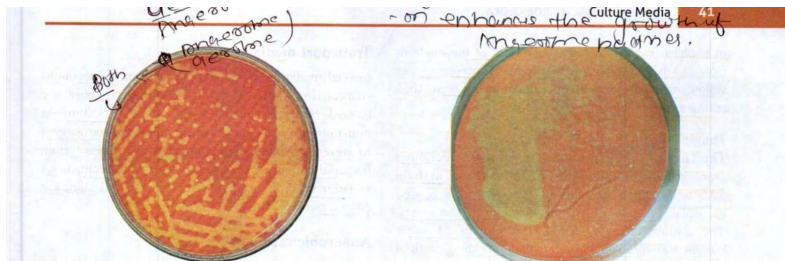


Fig. 4.1 Blood agar – *S. pyogenes*, β -haemolysis; Fig. 4.2 Blood agar – α -haemolytic viridans streptococci (from the book)



Fig. 4.3 TCBS agar – green colonies of *V. parahaemolyticus*; Fig. 4.4 Potassium tellurite agar – black colonies of *C. diphtheriae* (from the book)



Fig. 4.5 MacConkey agar – large mucoid *K. pneumoniae*; Fig. 4.6 MacConkey agar – flat pink *E. coli* (from the book)

Incubation: bacteria at **37°C for at least 48 h**; fungi at **22°C and 30°C**; both grow better in **5% CO₂**; anaerobic plates **7–14 days** in an anaerobic cabinet.

2. CULTURE METHODS

Reasons for pure culture: to identify organisms and their properties, test antibiotic sensitivity, study virulence and genetics, prepare antigens and vaccines, type isolates for epidemiology and archive for research.

Inoculation Methods

1. **Streak culture** – the routine method; a loopful (platinum/nichrome, 24 SWG, flamed and cooled) is spread on a dried plate in **parallel lines in successive segments**; well-separated colonies appear in the final streaks.
2. **Lawn (carpet) culture** – plate flooded with liquid culture for **uniform growth**; for **antibiotic sensitivity (disc method)**, phage typing and antigen preparation.
3. **Stroke culture** – on slopes, to obtain pure growth for tests.
4. **Stab culture** – long straight wire into a tube of medium; shows **oxygen requirement**, gelatin liquefaction; maintains stock cultures.
5. **Pour plate** – dilutions of 1 ml inoculum mixed with molten agar (**40–45°C**) and poured; colonies throughout the depth are counted – **viable count** (quantitative urine culture).
6. **Liquid culture** – large inocula (blood culture, sterility tests); dilutes antibiotics in the specimen; disadvantage – cannot separate mixed organisms.

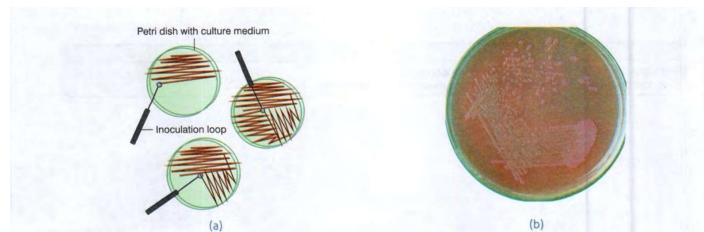


Fig. 5.1 Streak cultures (from the book)

Methods of Isolating Pure Cultures

- **Surface plating** – routine method giving distinct colonies.
- **Enrichment, selective and indicator media** – for specimens with mixed flora (feces).
- **Pretreatment** – e.g. sputum treated with **alkali/acid** (Petroff's) kills commensals but not tubercle bacilli.
- **Different temperatures** – only thermophiles grow at 60°C; at 22°C *N. catarrhalis* grows but not *N. meningitidis*.
- **Heating at 80°C** – kills vegetative forms, leaves spores (e.g. tetanus bacilli from dust).
- **Craigie's tube / U-tube** – semisolid agar; **motile** bacteria travel through the agar and are separated from non-motile ones; also for **phase variants**.
- **Animal inoculation** (anthrax bacilli from heart blood of mice) and **selective filters** (separating viruses from bacteria).

Anaerobic Culture Methods

Anaerobes vary in sensitivity to oxygen – strict (*C. tetani*) to aerotolerant (*C. histolyticum*). Anaerobiosis is achieved by exclusion/displacement of oxygen, vacuum, chemical or biological absorption of oxygen, and reducing agents.

1. McIntosh and Fildes' Anaerobic Jar

The **most reliable and widely used** method. A stout glass or metal jar with an airtight lid bearing **two taps (gas inlet and outlet)**, **two electric terminals** and a spool of **palladinised asbestos (catalyst)** under the lid. Plates are placed inside, air is **evacuated with a vacuum pump**, **hydrogen** is filled in, and the catalyst is heated electrically to combine hydrogen with residual oxygen to form water. Risk of explosion is avoided by using **room-temperature catalyst** (alumina pellets coated with palladium). **Indicator: reduced methylene blue** – colourless anaerobically, **blue with oxygen**. **Anoxomat** is an automated version.

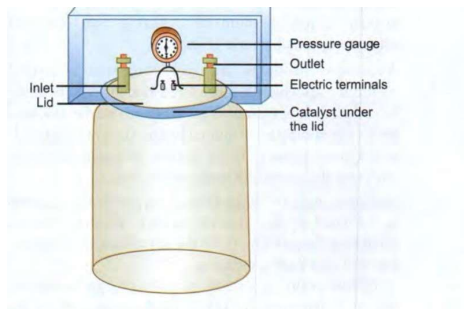


Fig. 5.2 McIntosh and Fildes' anaerobic jar (from the book)

2. Gaspak

Method of choice now. A disposable envelope containing chemicals that generate **hydrogen and CO₂** when **water is added**; placed in the jar with the plates and the lid closed. A **cold catalyst** combines hydrogen and oxygen. Simple, needs no vacuum or hydrogen supply.

3. Pre-reduced Anaerobically Sterilised (PRAS) media and 4. Anaerobic chamber (glove box)

PRAS media are commercially available for fastidious anaerobes. The **anaerobic glove box** is an airtight glass-fronted cabinet filled with inert gas, with an entry lock and gloves for handling.

5. Other Methods – Anaerobic Media and Reducing Agents

- **Robertson's cooked meat (RCM) medium** – fat-free minced cooked meat in broth; **most widely used fluid medium** for anaerobes; supports strict anaerobes; meat turns **red (saccharolytic)** or **black (proteolytic)**.
- **Thioglycollate broth** (with haemin and vitamin K) – reducing agent; **aerobes grow at the top, microaerophiles in the middle, anaerobes at the bottom**; boiled before use to expel oxygen; oxygen indicator turns pink/blue at the top.
- **Anaerobic broth with red-hot iron pieces** layered with vaseline; **Smith–Noguchi medium** (fresh animal tissue).
- **Reducing agents:** 1% glucose, **0.1% thioglycollate**, 0.1% ascorbic acid, 0.05% cysteine; a little agar slows oxygen diffusion.



Fig. 5.3 Growth in thioglycollate broth: (a) uniform turbidity (b) surface growth (c) granular turbidity (from the book)

3. BIOCHEMICAL TESTS

After isolation in pure culture, bacteria are identified by **morphology, staining, cultural characteristics, biochemical tests**, immunological methods and, now, automated systems, PCR and **MALDI-TOF**. Colonies are described by shape, size, elevation, margin, surface, colour, opacity and consistency.

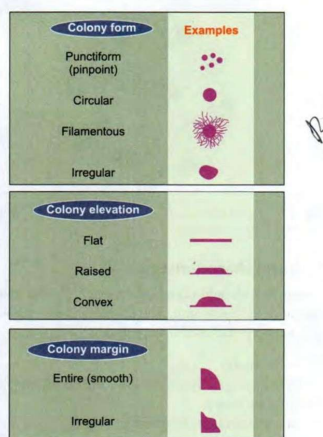


Fig. 6.5 Colony morphologic features (from the book)

Important Biochemical Tests

Test	Principle / medium & reagent	Positive result	Example (+)
Sugar fermentation	1% sugar in peptone water + indicator (Andrade's); Durham's tube for gas; Hiss' serum sugars (3% serum) for fastidious organisms	Acid – pink/red ; gas in Durham's tube	<i>E. coli</i> (glucose – acid + gas)
Indole	Tryptophan → indole ; peptone water, 48–96 h; Kovac's reagent	Red ring	<i>E. coli</i>
Methyl red (MR)	Mixed acid fermentation keeps pH < 4.5 ; glucose phosphate broth, 5 days at 30°C; methyl red	Red	<i>E. coli</i>
Voges–Proskauer (VP)	Acetyl methyl carbinol → diacetyl; 5% α-naphthol + 40% KOH	Pink/crimson in 2–5 min	<i>Klebsiella</i>
Citrate utilisation	Citrate as sole carbon source – Koser's citrate / Simmons' citrate	Turbidity (Koser's); blue (Simmons')	<i>Klebsiella</i>
Urease	Christensen's urease medium; urea → ammonia (alkaline)	Purple-pink	<i>Proteus</i> , <i>Klebsiella</i>
H₂S production	Sulphur amino acids → H ₂ S; lead acetate paper or ferric salts	Blackening	<i>Salmonella</i> , <i>Proteus</i>
Nitrate reduction	Nitrate reductase; 1% KNO ₃ broth; sulphanilic acid + α-naphthylamine	Red	Enterobacteriaceae
Catalase	H ₂ O ₂ → water + O ₂ ; H ₂ O ₂ on colony (not from blood agar)	Effervescence	Staphylococci (streptococci –ve)
Oxidase	Cytochrome oxidase; 1% tetramethyl-p-phenylenediamine ; plate or Kovac's filter paper	Purple/maroon within 10 s	<i>Pseudomonas</i> , <i>Vibrio</i> , <i>Neisseria</i>
Coagulase	Plasma clotting (slide – bound; tube – free)	Clot	<i>S. aureus</i>
Litmus milk	Acid, alkali, clot, peptonisation; stormy fermentation	Clot disrupted by gas	<i>C. perfringens</i>
Methylene blue reduction	1% methylene blue in broth culture	Decolourisation	–
Egg yolk (Nagler)	Lecithinase	Opacity/clearing zone round colonies	<i>C. perfringens</i>

Test	Principle / medium & reagent	Positive result	Example (+)
Bile esculin	Growth in 40% bile + esculin hydrolysis → black	Black	Enterococci
KCN test	Growth in potassium cyanide (1/13,000)	Growth	KCN-tolerant enterics

IMViC Tests

Indole, Methyl red, Voges–Proskauer and Citrate – used to identify **enteric Gram-negative bacilli**.

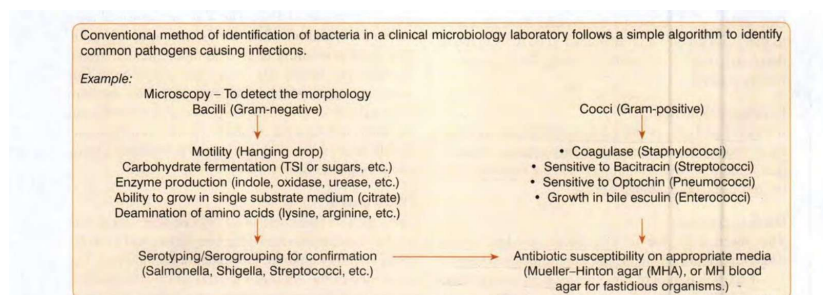
Organism	Indole	MR	VP	Citrate
<i>E. coli</i>	+	+	–	–
<i>Klebsiella pneumoniae</i>	–	–	+	+

Triple Sugar Iron (TSI) Agar – Composite Medium

Contains **glucose (0.1%)**, **lactose and sucrose (1% each)**, **phenol red** indicator and **ferrous sulphate** for H₂S, dispensed as **butt and slant**. It shows sugar fermentation, gas and H₂S in one tube – a preliminary identification of Gram-negative bacilli.

Slant / Butt	Interpretation	Example
Yellow / Yellow (A/A)	Glucose + lactose/sucrose fermented	<i>E. coli</i> , <i>Klebsiella</i>
Red / Yellow (K/A)	Glucose only fermented	<i>Shigella</i> , <i>Salmonella</i>
K/A + black butt	Glucose only + H ₂ S	<i>Salmonella</i> Typhi (small H ₂ S), <i>Proteus</i>
Red / Red (K/K)	No fermentation (non-fermenter)	<i>Pseudomonas</i>
Bubbles/cracks in butt	Gas production	<i>E. coli</i>

Identification Algorithm (from the book)



Conventional method of identification of bacteria in the clinical laboratory (from the book)

★ EXAM FOCUS

- Short notes: **Classification of culture media**, **Indicator media**, **Robertson's cooked meat medium**, **Gaspak**, **McIntosh–Fildes jar**, **Anaerobic culture methods**, **Craigie's tube**, **Methods of isolating pure cultures**.
- Short notes: **IMViC**, **TSI**, **Indole**, **Nitrate**, **Oxidase**, **Catalase**, **Urease**, **Citrate tests**.
- Remember: agar melts 98°C, sets 42°C; indicator in McIntosh jar – **reduced methylene blue**; Gaspak – method of choice.

